

Token-Efficient Multimodal Prototyping for Interpretable Cancer Survival Prediction

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Motivation

Cancer is a complex genetic disease with often unpredictable progression and has become one of the most pressing global health challenges. Accurate prognosis—particularly reliable survival analysis—is critical for clinicians and patients alike. Survival prediction functions as a form of patient risk assessment that enables precise stratification, guides the selection and intensity of treatment strategies, and helps patients and their families prepare emotionally and financially for future outcomes. Traditional approaches to survival prediction have included manual assessment, classical statistical models, and more recently, machine learning. Driven by the digitalisation of healthcare data, machine learning has emerged as a promising tool that avoids subjectivity and is capable of modelling complex non-linear relationships inherent in cancer progression. In particular, multimodal survival models that combine histology whole-slide images (WSIs) with gene expression profiles have become increasingly popular. These two data modalities provide complementary perspectives on tumour morphology and molecular alterations, mirroring the gold-standard assessments performed by pathologists in modern clinical practice. As a result, they consistently outperform single-modality approaches. Among these, early fusion models have been widely adopted due to their ability to capture cross-modal interactions and improve interpretability. In these models, WSI patches and gene groups are merged as tokens and processed by a Transformer to learn integrated patient representations. However, prior work has faced major limitations in handling the sheer scale of inputs: over 10^4 WSI patches and 20,000–30,000 gene expression features per patient. Processing such high-dimensional data without compromising intra-modality interactions or computational precision has remained challenging. To address this, we reproduce the multimodal prototyping (MMP) framework [1], which overcomes these constraints by aggregating both data modalities into concept-based prototypes, thereby reducing computational complexity while preserving both intra- and inter-modality relationships. Furthermore, we extend the original bimodal MMP by integrating a

third modality—DNA mutation data—which represents a primary driver of cancer development but has been underutilised due to its sparsity and high dimensionality. Finally, we visualise the bidirectional interactions among morphological, transcriptomic, and mutation-derived features, providing insights into latent patterns of cancer biology.

Methodology & Experiments

For histology data, we summarise WSI patches into 16 morphological prototypes with distinct tissue features using K-means clustering. Each WSI is then aggregated based on these prototypes with a Gaussian Mixture Model (GMM) to produce a patient-level embedding comprising 16 tokens. For genomic data, including both gene expression and mutation profiles, we employ the Hallmark Gene Sets as prototypes. Each set represents a defined cellular pathway with coherent expression patterns relevant to cancer survival. Only genes included in the Hallmark sets are retained and summarised into 50 pathway-level tokens. These tokens are fused and trained on downstream tasks using two approaches: (1) Transformer-based fusion and (2) optimal transport (OT) cross-alignment fusion. The integration of the third modality (mutation data) is applied only with the Transformer-based method, with mutation features weighted by their frequency and annotated impact to give the best accuracy. Experiments are conducted on The Cancer Genome Atlas (TCGA) across four cancer types (BRCA, LUAD, STAD, and KIRC). Model training uses 5-fold cross-validation with the Cox proportional hazards loss. We evaluate results against clinical, unimodal (MMP without data fusion), and multimodal (SurvPath [2]) baselines, reporting performance in terms of the C-index.

Key Findings

Both Transformer-based fusion and OT-based fusion within the MMP framework significantly outperformed clinical, unimodal, and other multimodal baselines. Transformer-based fusion achieved the highest average C-index of 0.685, demonstrating the strong predictive capability of MMP. These superior results reflect the use of high-quality prototyping, which effectively com-

bines complementary information from multiple modalities, reduces computational demands, and enables simultaneous modelling of intra- and inter-modality interactions. At the cohort level, except for KIRC—where unimodal histology-based prediction achieved the highest accuracy—all other cancer types showed the best performance with MMP models. The performance difference between the Transformer-based and OT-based approaches was relatively small, with the Transformer variant achieving slightly higher accuracy (+0.9%). This likely results from the greater stability of Transformers trained with backpropagation and residual connections, compared to OT methods relying on iterative Sinkhorn optimisation. In addition, the MMP models demonstrated statistically significant risk stratification in log-rank tests. Integrating the third modality (mutation data) further increased the C-index to 0.694. Although this improvement was modest, it highlights the value of incorporating diverse data modalities. It also suggests partial redundancy in mutation information and limitations in using hallmark gene sets for prototyping, which may exclude key mutated genes not directly related to canonical cellular processes. Finally, we visualised bidirectional relationships between modalities by interpreting cross-modal attention scores. For selected histology prototypes, we identified the most correlated gene expression pathways and vice versa. These insights have great potential for uncovering hidden genetic mechanisms of cancer, informing targeted therapy development, and clarifying how gene expression alterations manifest at the tissue level.

Conclusion & Next Steps

This project confirmed that multimodal prototyping enables accurate and interpretable cancer survival prediction while substantially reducing computational cost. Incorporating mutation data further improved predictive performance, but results also revealed important limitations.

Future directions include:

- Developing data-driven methods to define prototypes dynamically.
- Integrating cancer-type-specific knowledge to better capture relevant genomic alterations.
- Extending the framework to other prognostic tasks, such as recurrence and treatment response prediction.

Overall, this work provides a foundation for more scalable and explainable survival prediction mod-

els that can support personalised, data-driven cancer care.

References

- [1] Andrew H. Song, Richard J. Chen, Guillaume Jaume, Anurag Vaidya, Alexander S. Baras, and Faisal Mahmood. Multimodal prototyping for cancer survival prediction. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *PMLR*, Vienna, Austria, 2024. PMLR.
- [2] Guillaume Jaume, Anurag Vaidya, Richard J. Chen, Drew F. K. Williamson, Paul Pu Liang, and Faisal Mahmood. Modeling dense multimodal interactions between biological pathways and histology for survival prediction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11579–11590, 2024.

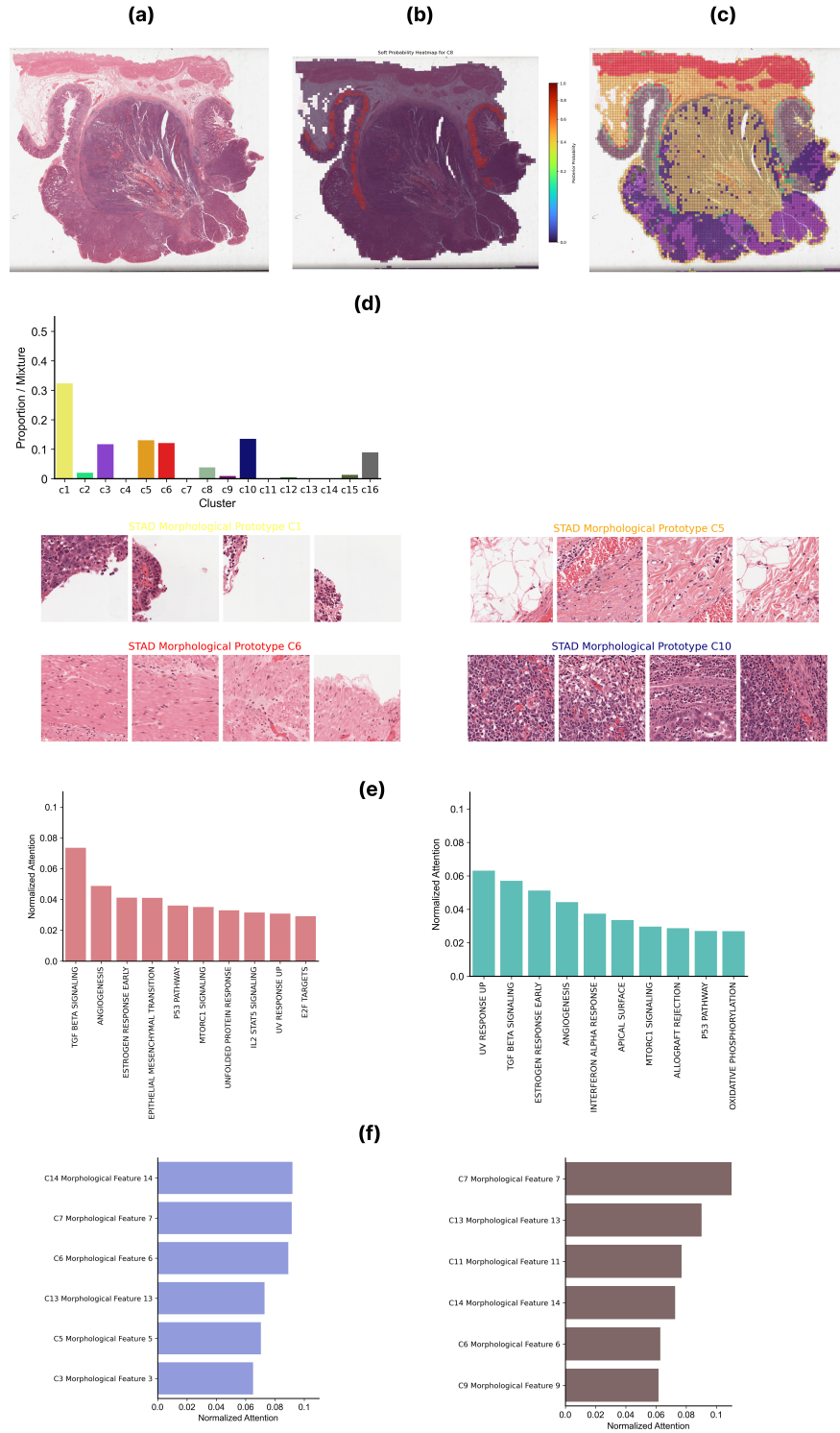


Figure 1: Visualisations of STAD cohort results using the MMP framework. (a) Original whole-slide image. (b) Soft probability heatmap. (c) Prototype assignments. (d) Proportion distributions and examples. (e) Top attention weights. (f) Top morphological attentions.